OpenTelemetry: Playtime is Over

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Ahoj!

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Download PDF of the slides here!

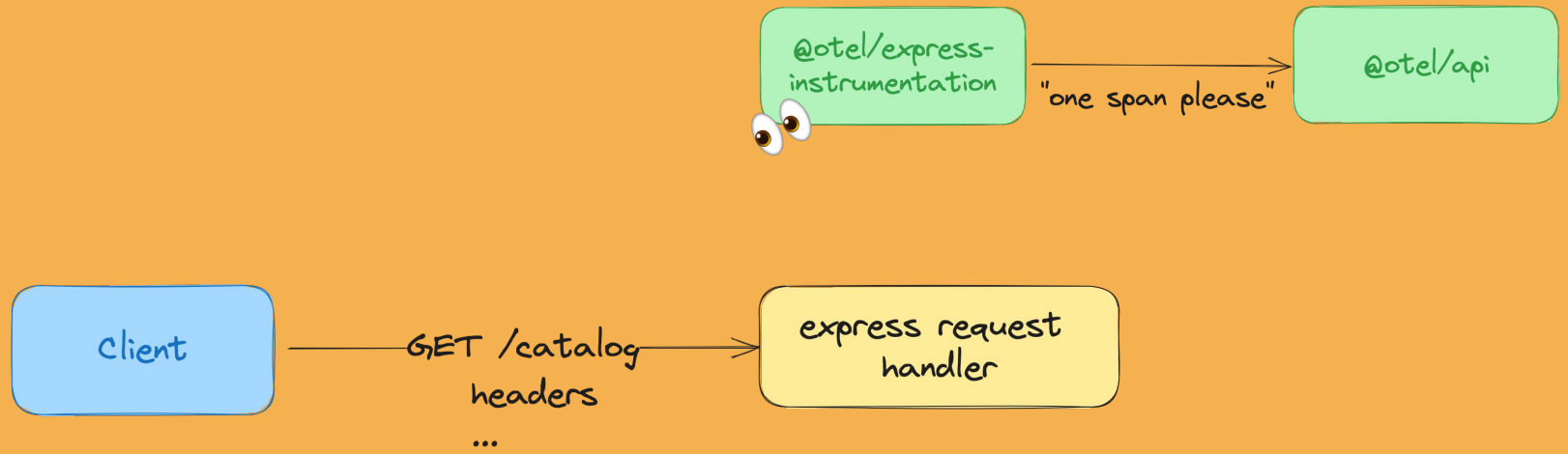
01

We’re past the OTel honeymoon

Time to grow up

- Managing costs
- Telemetry quality
- Collector fleet management

The easy path isn't always the best path



02

Cutting down the noise (and costs)

PROBLEM: Information overload

Intentional instrumentation

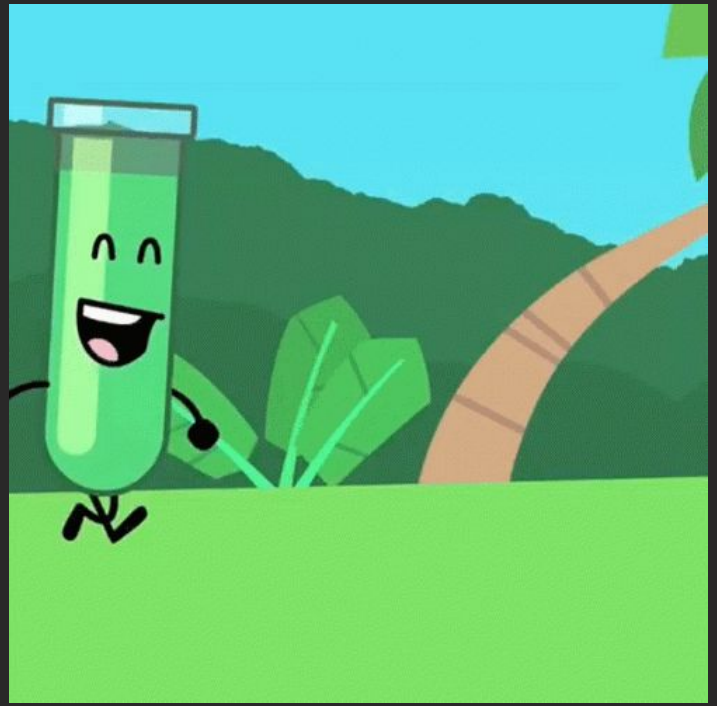
- 🔭 Built-in is best 😎
- 🔭 Library instrumentation: 👍
- 🔭 No-code instrumentation: 😐
- 🔭 Standards: Weaver, MCP 😉

Auto-instrumentation gotchas

- Lots of spans
- Can be configured*
- Computational overhead*
- Filter as soon as possible

Sampling

- We don’t care about every HTTP 200 response
- Reduce costs without losing visibility
- Filter first, then sample
- Head vs Tail sampling



Metrics

- Metrics are deceptively complex
- Cardinality, schmardinality
- Don’t scrape what you don’t need
- Target allocator is 🙌

03

Pipeline efficiency

PROBLEM: Telemetry is expensive!

OTLP protobuf ain’t perfect

Hierarchical telemetry data
Repetitive metadata

OTLP is row-based
Inefficient for storage

Serialize/deserialize
Encoding/decoding gets expensive

CNCF + Apache = OTel Arrow ❤️

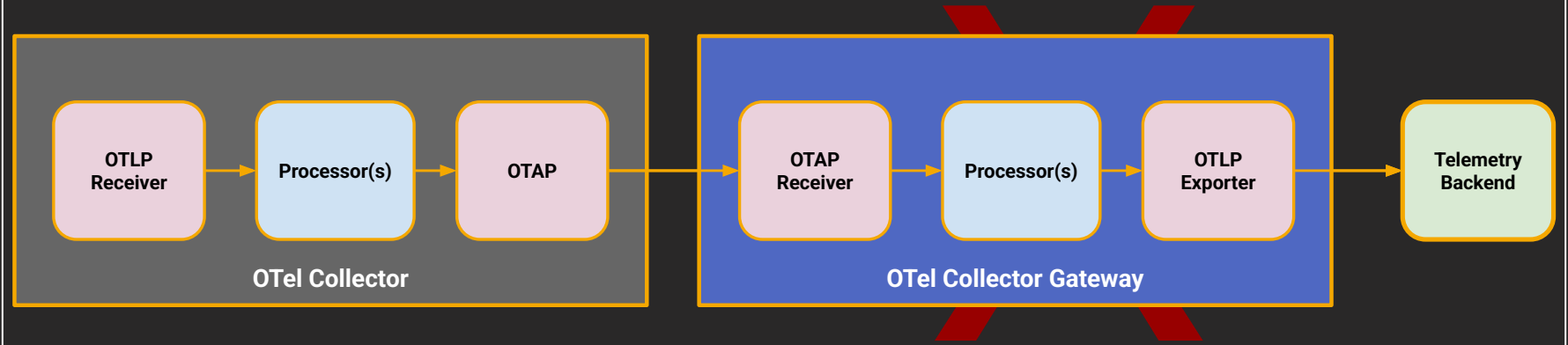
OTel Arrow is based on the Apache Arrow framework, and serves both a data format and a transport protocol.

OpenTelemetry + APACHE ARROW

OTel Arrow

- Columnar format
- Zero-copy data sharing
- Available as Collector components

Current State



Next phase

Telemetry Data Source

Telemetry Backend

04

Schema management & validation

PROBLEM: Managing semantic convention drift

OTel Weaver to the Rescue!

A tool for creating, documenting, managing, and validating OpenTelemetry schemas and policies.

How it works

Create Schema (YAML) → Create Templates (Jinja2) → Generate Code from Templates → Validate in CI

Requires learning Rego & Jinja2

Weaver Overview →

05

Collector Management

PROBLEM: Oh so many Collectors to configure

OpAMP to the rescue!

Remote management of data collection agents!

OpAMP Overview

OpAMP client & server

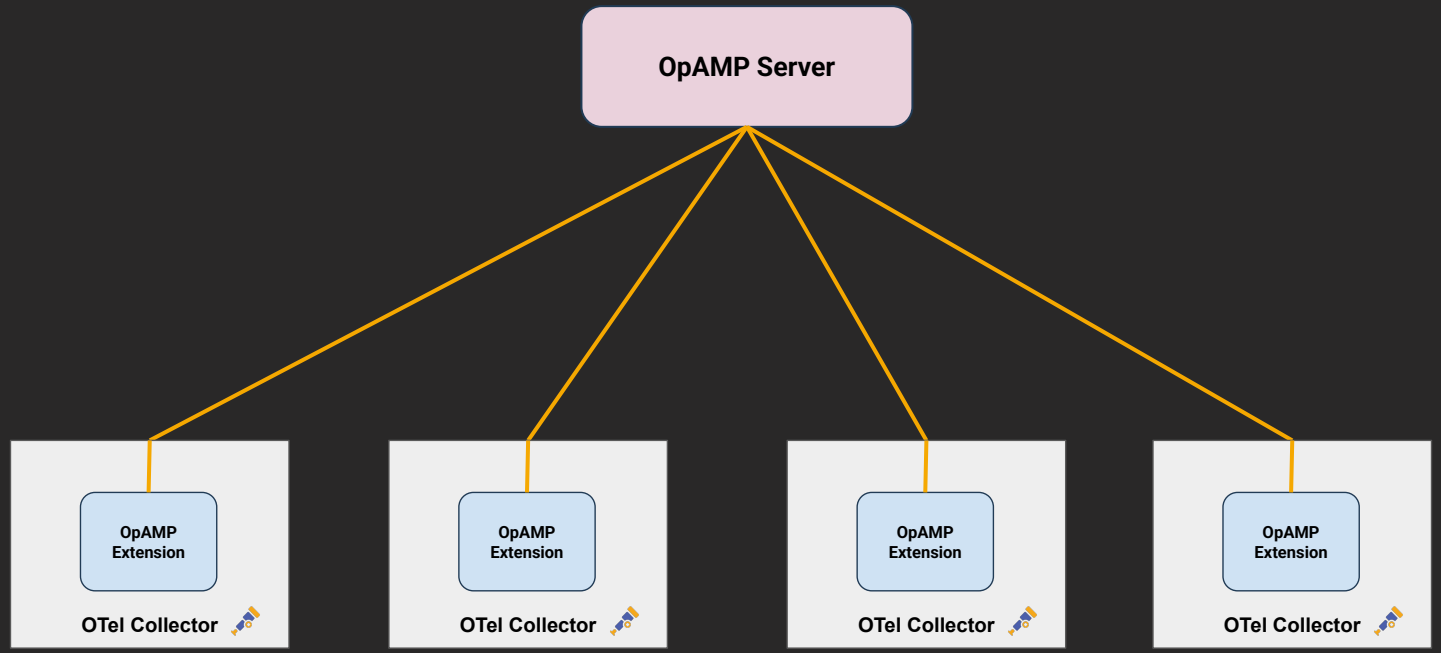
OpAMP Server
Sends config updates, packages, instructions

OpAMP Client
Reports status, applies updates, receives configs

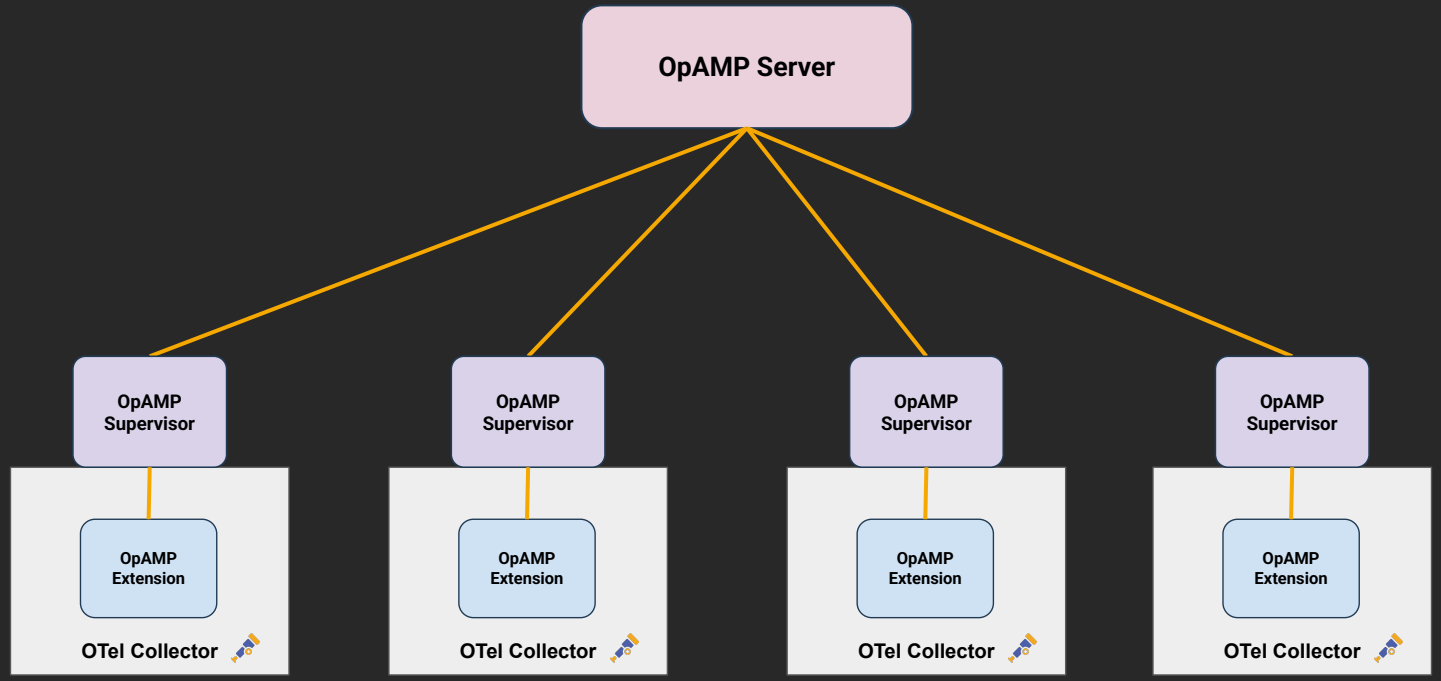
Choose from existing server or write your own!

OpAMP on the Collector?

OpAMP Extension



OpAMP Supervisor



OTel tools gotchas

- Many moving parts
- Large learning curve
- Sparsely documented

06

Now WHAT?

What you can do NOW!

- Look at areas for improvement
- Use AI agents to stay up to date
- Don’t boil the ocean
- Ask for help

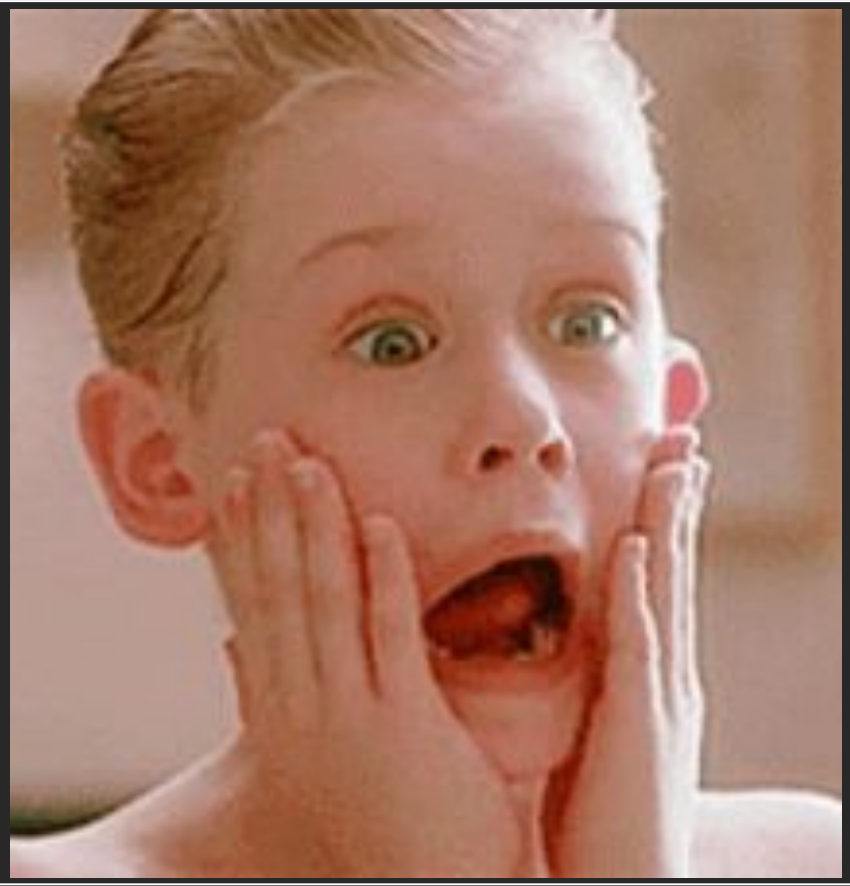
OTel Maturity Matrix (#goals)

Basics	Good	Better	At Scale
• RED Metrics • Resource utilization • Logging with levels • Black-box testing	• Tracing for key paths • Monitoring for your monitoring • Custom dimensions / tags • SLOs	• Prevalent tracing • Instrumented internal libraries • Internal sem convs and docs with Weaver	• Optimized pipelines with Arrow • Fleet management with OpAMP • Weaver live-checking

Don’t Be Tempted by…

- Auto-instrumentation overload
- Tracing as an afterthought
- Instrumentation outsourcing
- Latting AI take the wheel

Don’t panic!



Díky!

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Slides & resources

BONUS:
Collector-ing “well”

BYOC (Build Your Own Collector)

- Most people use Collector contrib
- Contrib == bloat + insecure
- Build your own Collector
- Use OTel Collector builder

Collector Deployment Patterns

- Gateway(s) + sidecars
- Collector per signal
- StatefulSet
- No Collector (serverless)

Audits

- Coverage
- Quality
- Usefulness
- Pipeline Health
- Governance